

Matching 2D Image Patches and 3D Point Cloud Volumes by Learning Local Cross-domain Feature Descriptors

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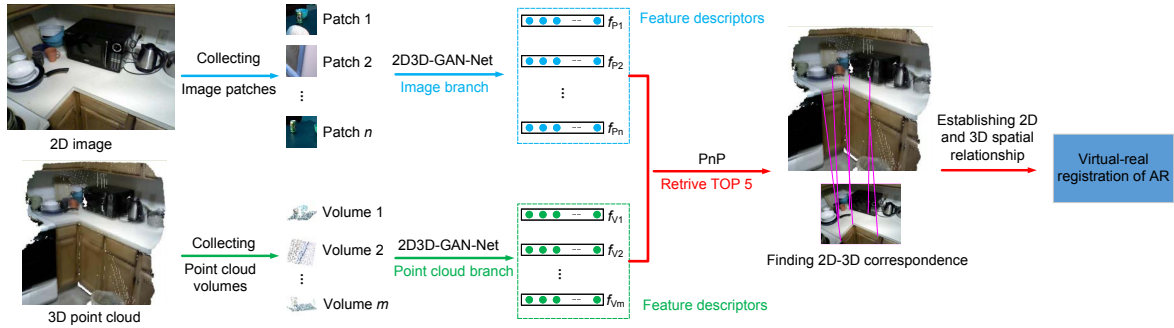


Figure 1: The pipeline of proposed AR virtual-real registration by using the local cross-domain feature descriptors learned by 2D3D-GAN-Net. 1) Cropping 2D image patches and 3D point cloud volumes by using the 2D and 3D keypoints. 2) Using 2D3D-GAN-Net to extract the local cross-domain feature descriptors f_{p_n} and f_{v_m} . 3) Retrieving the TOP5 results from each 2D feature descriptors to the 3D feature descriptors database, and using Perspective-n-Point (PnP) algorithm to find the 2D-3D correspondences to calculate the camera pose. 4) Using the camera pose to compute the transformation matrix to achieve virtual-real registration of AR.

ABSTRACT

Establishing the relationship of 2D images and 3D point clouds is a solution to establish the spatial relationship between 2D and 3D space, i.e. AR virtual-real registration. In this paper, we propose a network, 2D3D-GAN-Net, to learn the local invariant cross-domain feature descriptors of 2D image patches and 3D point cloud volumes. Then, the learned local invariant cross-domain feature descriptors are used for matching 2D images and 3D point clouds. The Generative Adversarial Networks (GAN) is embedded into the 2D3D-GAN-Net, which is used to distinguish the source of the learned feature descriptors, facilitating the extraction of invariant local cross-domain feature descriptors. Experiments show that the local cross-domain feature descriptors learned by 2D3D-GAN-Net are robust, and can be used for cross-dimensional retrieval on the 2D image patches and 3D point cloud volumes dataset. In addition, the learned 3D feature descriptors are used to register the point cloud for demonstrating the robustness of learned local cross-domain feature descriptors.

Index Terms: Human-centered computing—Human Computer Interaction—Interaction Paradigms—Augmented reality

1 INTRODUCTION

3D point clouds provide potential basic 3D information support for AR applications. Essentially, if the 2D images and the 3D point clouds can be matched, the spatial relationship of 2D and 3D space will be established, which is a solution to achieve the virtual-real registration of AR [2]. However, it is challenging to match the 2D

images and 3D point clouds, because the definitions of handcrafted keypoints or feature descriptors for 2D images and 3D point clouds are extremely different. The goal of our work focuses on learning the invariant local cross-domain feature descriptors of 2D image patches and 3D point cloud volumes, which can be used to match the 2D images and 3D point clouds. The pipeline of the proposed AR virtual-real registration approach in this paper is shown in Figure 1.

In this paper, we propose a novel network framework, 2D3D-GAN-Net, to learn the local invariant cross-domain feature descriptors. The embedded GAN, which forces 2D3D-GAN-Net to correctly distinguish which domain data the learned feature descriptors come from, facilitates the extraction of invariant local cross-domain feature descriptors. The local cross-domain feature descriptors of 2D image patches and 3D point cloud volumes learned from 2D3D-GAN-Net achieve cross-dimensional retrieval performance (2D to 3D or 3D to 2D) on the 2D image patches and 3D point cloud volumes dataset.

2 METHOD

As shown in Figure 2, the proposed 2D3D-GAN-Net consists a Generator G and a Discriminator D . 2D3D-GAN-Net aims to use an adversarial strategy to encourage the paired matching 2D and 3D feature descriptors learned by the G to be as similar as possible, and let the D to distinguish the source of the learned feature descriptors from which domain data. The inputs of 2D3D-GAN-Net are paired matching 2D image patches (size: $64 \times 64 \times 3$) and 3D point cloud volumes (size: 1024×6). The outputs of 2D3D-GAN-Net are the 2D and 3D feature descriptors, which are 256-dimensional vectors.

The 2D3D-GAN-Net is optimized with the interplay between the G and D by using Binary Cross Entropy (BCE), defined as:

$$L_D = L_{BCE}(D_\beta(G_\theta(I)), 1) + L_{BCE}(D_\beta(G_\theta(P)), 0) \quad (1)$$

$$L_G = \lambda_1 (L_{BCE}(D_\beta(G_\theta(I), 0)) + L_{BCE}(D_\beta(G_\theta(P), 1))) + \lambda_2 L_{Hard} \quad (2)$$

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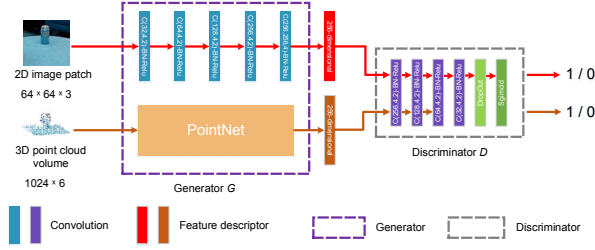


Figure 2: The architecture of 2D3D-GAN-Net.

	2D3D-GAN-Net	LCD	2D3D-MatchNet	Siam2D3D-Net
TOP1	0.5842	0.5778	0.1520	0.1412
TOP5	0.8811	0.8624	0.3897	0.3569

Table 1: The TOP1 and TOP5 retrieval accuracy of proposed 2D3D-GAN-Net and comparative networks.

where, I and P are the inputs of paired matching 2D image patches and 3D point cloud volumes; G and D are the generator and discriminator, respectively; θ and β are the parameters of G and D , respectively; label 1 denotes that the f_I learned from I and the f_P learned from P ; label 0 denotes f_I learned from P and the f_P learned from I ; λ_1 and λ_2 are the weights of BCE and hard triplet losses, respectively. L_{Hard} is a triplet loss to make matching samples closer and make non-matching samples separate, defined as:

$$L_{Hard} = \frac{1}{n} \sum_{i=1, n} \max(0, 1 + d(i_i, p_i) - \min(d(i_i, p_{j_{min}}), d(c_{k_{min}}, p_i))) \quad (3)$$

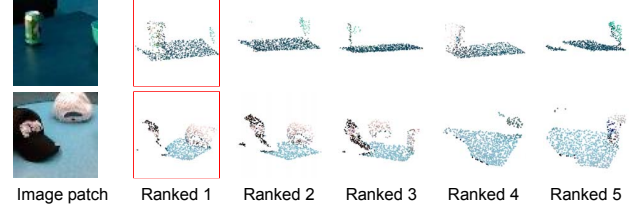
where, i_i and p_i are the closest feature descriptors distance for matching pair I_i and P_i , the closest non-matching feature descriptor for i_i is $p_{j_{min}}$ and the closest non-matching descriptor for p_i is $i_{k_{min}}$.

The 2D3D-GAN-Net is implemented with PyTorch framework, and trained it with a Nvidia 3090 Ti GPU. Both discriminator and generator are trained with the SGD optimizer, the momentum is set as 0.9. Initially, the learning rates for the generator and discriminator are set at 0.001. The learning rate of generator decreased 0.0005 after each epoch, and the generator decreased 0.0005 after 5 epochs.

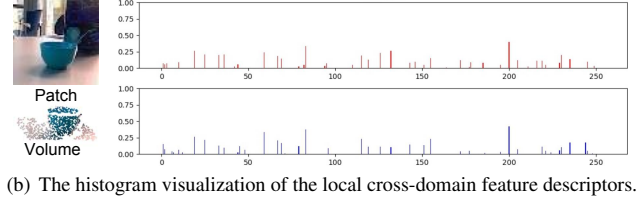
3 EXPERIMENT

The 2D image patches and 3D point cloud volumes used in this paper is constructed from the 3DMatch dataset [4]. 580,000 pairs of matching samples are used for training, and 20,000 pairs of matching samples are used for testing. They do not intersect each other.

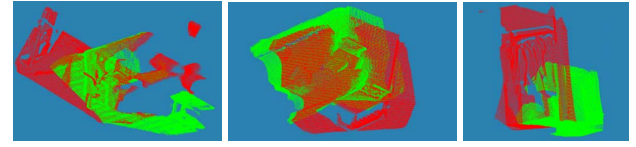
Compare with existing best-performing methods LCD [3], 2D3D-MatchNet [1] and Siam2D3D-Net [2], the cross-domain feature descriptors learned by 2D3D-GAN-Net achieves state-of-the-art retrieval performance from 2D image patches to 3D point cloud volumes on the testing dataset, shown in Table 1. For fair comparison, 2D3D-GAN-Net and all competing networks are trained for 20 epochs. Several TOP5 retrieved ranking results of 2D image patches retrieve to 3D point cloud volumes are shown in Figure 3(a), the true matching results are labeled with red bounding boxes. The histogram visualization of the cross-domain feature descriptors learned by 2D3D-GAN-Net are visualized in Figure 3(b). It can be viewed that the distribution of the histogram are similar, which demonstrates that the local cross-domain feature descriptors learned by 2D3D-GAN-Net are robust. Finally, several point cloud registration in indoor scene by using the 3D feature descriptors learned from the proposed 2D3D-GAN-Net demonstrate the applicability of learned 3D feature descriptors, as shown in Figure 3(c).



(a) The retrieval TOP5 ranking results.



(b) The histogram visualization of the local cross-domain feature descriptors.



(c) The registration results of the source and target 3D point cloud.

Figure 3: The retrieval TOP5 ranking results, histogram visualization and registration results by using the local cross-domain feature descriptors learned by our proposed 2D3D-GAN-Net.

4 DISCUSSION

Compared with the traditional 2D and 3D feature descriptors, 2D3D-GAN-Net benefit with directly processing the raw iamges and point clouds without prior information to learn invariant local 2D and 3D feature descriptors. However, the obstructed, incomplete point clouds, and inconsistent lighting images will cause great interference to 2D3D-GAN-Net, resulting in unrobust extracted features. In addition, we also tested the case without adversarial loss, and its TOP1 retrieval accuracy dropped to 0.3045, which demonstrates that adversarial loss plays an important role in 2D3D-GAN-Net.

5 CONCLUSION

Our proposed 2D3D-GAN-Net is an exploratory attempt to jointly learn local cross-domain feature descriptors of the 2D image patches and 3D point cloud volumes. The matched 2D image patches and 3D point cloud volumes by using the feature descriptors learned from 2D3D-GAN-Net have the potential to construct 2D and 3D spatial relationship. So that our exploration work have the potential to support the basic work of virtual-real registration of AR.

6 ACKNOWLEDGMENTS

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